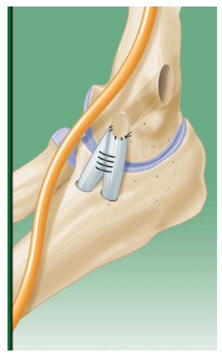

Elbow Diagnosis and Decision-Making

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OVERVIEW OF PATHOLOGIES

The key to diagnosing elbow injuries involves performing a thorough history and physical examination of the elbow. Additionally, to rule out referred pain, the surrounding areas such as the cervical spine, shoulder, wrist, and hand should also be examined. This chapter discusses the elements of a complete elbow history and physical examination, which would include inspection, palpation, range of motion (ROM), strength, neurologic assessment, joint stability testing, and various provocative maneuvers to help identify and characterize different disorders around the athlete's elbow.

Owing to the wide variety of elbow pathologies, a systematic approach is necessary to diagnose the disorders accurately. A helpful strategy is to divide the elbow into four anatomic regions: lateral, medial, anterior, and posterior. This strategy enables the examiner to narrow the range of the differential diagnosis. In addition, knowing the patient's athletic history and any previous history of injury can help focus decision-making regarding treatment.

Pathologic conditions that are present around the lateral aspect of the elbow include lateral epicondylitis, radial nerve compression neuropathies, osteochondritis dissecans (OCD) of the capitellum, posterolateral rotatory instability, and radiocapitellar arthritis. Symptoms arising from the medial aspect of the elbow include medial epicondylitis, cubital tunnel syndrome, flexor-pronator strain, and medial (ulnar) collateral ligament injury. The differential diagnoses for anterior elbow pain include median nerve compression neuropathies, distal biceps tendinopathy, partial or complete distal biceps tendon ruptures, and an anterior capsular strain from a hyperextension injury. Conditions causing symptoms at the posterior elbow include olecranon bursitis; olecranon stress fracture; posteromedial impingement syndrome, such as a valgus extension overload as seen in overhead-throwing athletes; triceps tendinopathy; or a triceps rupture.

HISTORY

Athletes typically present with pain, mechanical symptoms, and a compromised ability to play. The pain may be exacerbated by activity and relieved with rest, or it can be persistent with daily routines. It is therefore helpful to localize the athlete's symptoms into one of the four anatomic regions previously described.

Lateral Elbow

Symptoms involving the lateral elbow are typically related to lateral epicondylitis, radial nerve compression neuropathies, OCD lesion of the capitellum, posterolateral rotatory instability, and/or radiocapitellar arthritis.

Lateral epicondylitis, often known as "tennis elbow," is caused by tendinosis and inflammation at the origin of the extensor carpi radialis brevis (ECRB). Patients often present with a "burning" pain that radiates from the lateral epicondyle distally along the extensor muscles of the forearm. This pain is intensified by a combination of elbow extension, resisted wrist extension, and a tight grip. Eccentric loading during these combined maneuvers, as seen when hitting a backhand late in tennis or catching a golf swing "fat" (i.e., striking the ground before the ball), may trigger the symptoms associated with lateral epicondylitis.

Sharp pain in the lateral aspect of the elbow associated with locking or catching can result from loose bodies in the radiocapitellar joint or an OCD lesion of the capitellum. Patients at risk for OCD lesions include young overhead throwers and gymnasts. Young overhead throwers develop OCD lesions as an overuse injury from increased compression and shear forces during the late cocking phase of throwing across the radiocapitellar articulation. The radiocapitellar joint also experiences 60% of axial load placed across the elbow, which can cause OCD lesions in gymnasts after repetitive loading.[1] These patients often report an insidious onset of symptoms with limitation of elbow motion, particularly elbow extension.[2] Mechanical symptoms involving the elbow are often found in patients with radiocapitellar arthritis, with symptoms most severe at extremes of elbow motion.

Radial nerve compression syndromes include compression neuropathies involving the posterior interosseous nerve (PIN) and radial tunnel syndrome (RTS). PIN syndrome is characterized by a pure motor loss of extension of the thumb and fingers. This syndrome is exceptionally uncommon in athletes in the absence of a mass in the region of the radial neck, prolonged compression as seen in the "Saturday night palsy," or Parsonage-Turner syndrome. RTS is also quite uncommon. It is characterized by pain that is exacerbated with activity and commonly located in the extensor mobile wad and proximal lateral forearm. Throwing athletes may experience this pain at the end of follow through. Symptoms of posterolateral rotatory instability usually follow a

Abstract

The key to diagnosing elbow injuries involves performing a thorough history and physical examination of the elbow. This chapter discusses the elements of a complete elbow history and physical examination, including inspection, palpation, range of motion, strength, neurologic assessment, joint stability testing, and various provocative maneuvers to help identify and characterize the disorders that may involve an athlete's elbow. owing to the wide variety of elbow pathologies, a systematic approach is necessary to arrive at an accurate diagnosis. A helpful strategy is to divide the elbow into four anatomic regions: lateral, medial, anterior, and posterior. This strategy enables the examiner to narrow the range of the differential diagnosis. Pathologic conditions that present from the lateral of the elbow include lateral epicondylitis, radial nerve compression neuropathies, osteochondritis dissecans (OCD) of the capitellum, posterolateral rotatory instability, and radiocapitellar arthritis. Symptoms arising from the medial aspect of the elbow include medial epicondylitis, cubital tunnel syndrome, flexor-pronator strain, and medial (ulnar) collateral ligament injury. The differential diagnoses for anterior elbow pain include medial nerve compression neuropathies, distal biceps tendinopathy, partial or complete distal biceps tendon ruptures, and an anterior capsular strain from a hyperextension injury. Conditions causing symptoms at the posterior elbow include olecranon bursitis, olecranon stress fracture, posteromedial impingement syndrome—such as a valgus extension overload, as seen in overhead-throwing athletes—triceps tendinopathy, or a triceps rupture.

Keywords

elbow
decision-making
diagnosis
physical examination

history of prior elbow dislocation in an athlete. Complaints include lateral-sided elbow pain, snapping, locking, subjective instability, and recurrent elbow dislocation.

Medial Elbow

Complaints localized to the medial aspect of the elbow may result from cubital tunnel syndrome, medial epicondylitis, and medial (ulnar) collateral ligament injury.

Paresthesias radiating along the medial portion of the forearm to the ring and small finger are typical of cubital tunnel syndrome. Patients often complain of nocturnal symptoms or symptoms exacerbated by prolonged elbow flexion. Popping or snapping along the medial aspect of the elbow may result from a subluxating ulnar nerve or movement of the medial head of the triceps after ulnar nerve transposition.

Medial epicondylitis, also known as "golfer's elbow," is also found in racquet sports and baseball. It is associated with a history of repetitive forceful forearm pronation or wrist flexion. In golfers, it typically involves the dominant arm. A single "fat" shot leading to eccentric loading of the flexor-pronator mass or repetitive loading of the elbow can lead to disruption of the flexor pronator origin, with ischemia and development of angiofibroblastic hyperplasia.[3] Medial epicondylitis has been associated with ulnar nerve symptoms in up to 60% of cases.[4]

Athletes who throw and sustain an isolated acute medial collateral ligament injury will frequently describe a pop, sharp pain, and the inability to continue pitching or throwing. Insidious medial elbow pain with associated loss of velocity and control in throwing athletes are signs of chronic medial collateral ligament injury. Medial elbow discomfort in the setting of acute elbow trauma from higher-energy mechanisms can relate to disruption of the medial collateral ligament. These patients typically have associated injuries, such as radial head fracture, capitellum fracture, or dislocation. Throwing athletes with flexor-pronator strains have a history similar to that of medial collateral ligament injury.

Anterior Elbow

Athletes may present with anterior elbow pain for a number of reasons, including partial or complete distal biceps rupture, distal biceps tendinosis, median nerve compression neuropathies such as pronator or anterior interosseous nerve (AIN) syndrome, and anterior capsular strains.

Distal biceps tendon ruptures are most often reported with a traumatic event involving an unexpected extension force applied to the flexed arm that can occur in weight lifting, rugby, or football. A history of anabolic steroid use is helpful, as this is a significant risk factor for tendon ruptures, most notably in the upper extremity.[5] A distal biceps rupture is usually associated with a pop followed by pain and weakness in the upper extremity. The intense pain recedes quickly, as does the initial weakness of elbow flexion. Patients are usually left with weakness in forearm supination and, rarely, cramping in the arm. Distal biceps tendinosis is associated with anterior elbow pain that is more pronounced with repetitive forearm supination and resisted elbow flexion.

Pronator syndrome is a rare condition that results most commonly from compression of the median nerve as it passes between the two heads of the pronator teres. There are other locations of potential compression; these are described later in the chapter. Patients present with aching pain in the proximal volar forearm and paresthesias radiating into the radial side of the hand, namely the thumb to the radial border of the ring finger. Symptoms may worsen with repetitive forearm rotation as well. AIN syndrome is a pure motor palsy with symptoms found distally in the hand. However, patients may have a vague achy discomfort in the anterior elbow associated with median nerve compression at that level.

Posterior Elbow

Athletes may experience posterior elbow pain due to olecranon bursitis, a stress fracture, posteromedial impingement syndrome (e.g., valgus extension overload syndrome), triceps tendinosis, and triceps tendon rupture. Olecranon bursitis typically occurs after either a single traumatic episode or with repetitive pressure or a shearing force on the apex of the elbow. Olecranon stress fractures are seen most often in gymnasts, athletes who throw, and weight lifters (Fig. 58.1). These athletes present with reports of posterior elbow pain, loss of terminal extension, and in some cases elbow clicking or catching.

Posteromedial impingement syndrome (e.g., valgus extension overload syndrome) is nearly exclusive to throwing athletes who

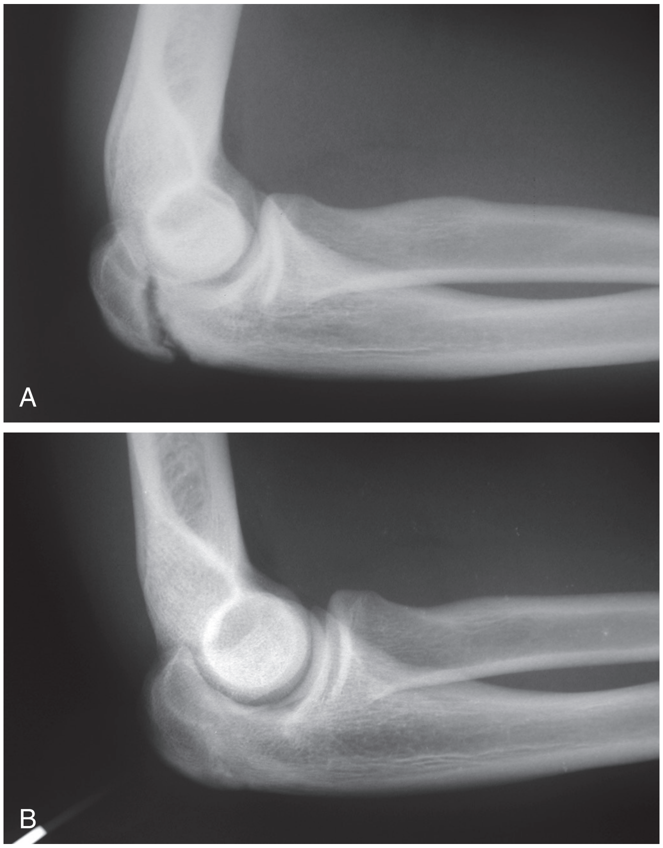

Fig. 58.1 An olecranon stress fracture. Radiographs of a college gymnast who presented with posterior elbow pain are shown. (A) The symptomatic side. (B) The contralateral extremity is shown for comparison.

report pain at the posteromedial aspect of the elbow during the follow-through phase while reaching terminal elbow extension. This is usually secondary to posterior olecranon osteophytes impinging in the fossa at the terminal extension phase.

Patients who have triceps tendinopathy have pain with elbow extension, especially against resistance. An athlete with a distal triceps tendon rupture will often recall a specific traumatic event during resisted elbow extension, followed by intense pain and significant weakness.[6] Rupture may occur with minimal trauma in patients who use anabolic steroids.[5]

Overall, elbow injuries are most commonly found in throwing athletes. The American Academy of Orthopaedic Surgeons (AAOS) reports that 50% to 75% of adolescent baseball players report episodes of elbow pain during play. In taking a history, especially from a throwing athlete, it is important to determine when symptoms occur in the throwing phase. Medial-sided symptoms usually occur during the late acceleration and cocking phase compared with posterior elbow symptoms, which occur during the deceleration and follow-through phase.

PHYSICAL EXAMINATION

For a diagnostic test to be clinically useful, it must be easily performed, reliable, reproducible, and have high sensitivity and specificity. In evaluating elbow injuries, a combination of inspection, palpation, ROM, strength, stability, neurologic assessment, and other provocative maneuvers will assist in the development of a working diagnosis.

Inspection

Inspection is the initial step of the physical examination of the elbow and, if possible, should include a comparison of the injured side with the uninjured side. During inspection, the physician should assess the skin and soft tissue envelope for signs of acute injury such as ecchymosis, swelling, deformity, abnormal muscle contour, and chronic changes such as muscle hypertrophy or atrophy.

During inspection, a measurement of the carrying angle is useful, especially for throwing athletes. The carrying angle of the elbow is the angle between the axis of the humerus and the axis of the forearm measured with the elbow extended and the forearm maximally supinated (the anatomic position) in the coronal plane. Typically, the angle is 7 to 13 degrees of valgus.[7] Paraskevas et al.[8] found that the mean carrying angle of the general population is almost 13 degrees, with men averaging 2 degrees less and women 2 degrees more. The increased angle in women has been attributed to increased joint laxity in women compared with men. Variations in the carrying angle can be due to previous trauma, injury, developmental abnormality, or adaptive changes to repetitive stress.[8,9] In professional throwers, it is common to see carrying angles greater than 15 degrees.[10] Fick submitted the "muscle theory," which accounts for the greater carrying angle in athletes who throw. This theory hypothesizes that the increased strength of the brachioradialis and extensor carpi radialis longus in athletes who throw will cause a greater radial deviation in extension and thus a greater carrying angle. It has also been hypothesized that an increased amount of stress on the elbow joint in athletes who throw may lead to increased laxity of the joint, which also contributes to a greater carrying angle.

When evaluating patients for atrophy or hypertrophy of muscles of the arm or forearm, girth measurements compared with the contralateral side are useful when appropriate. Hypertrophy of the forearm musculature is often present in the dominant extremity of many types of athletes. Atrophy of the arm and forearm muscle may also indicate disuse because of chronic discomfort or an underlying neurologic disorder.

Palpation

It is important to start with palpation of asymptomatic areas first while progressively moving toward the symptomatic area. Additionally, the physician must understand the qualitative aspect of this portion of the examination and use maneuvers while palpating to understand what exacerbates or relieves symptoms.

Lateral Elbow

Beginning with the lateral elbow, the bony prominences should be palpated, including the lateral epicondyle, radial head, and olecranon process. The radiocapitellar joint line should also be palpated for tenderness. Based on the history, point tenderness of the lateral epicondyle in the setting of trauma can be indicative of lateral collateral ligament disruption.

Tenderness due to lateral epicondylitis may be palpated just anterior and distal to the lateral epicondyle at the ECRB origination and may extend distally from the lateral epicondyle for several centimeters along the common extensor origin. Pain associated with lateral epicondylitis is generally worsened by elbow extension with resisted wrist extension and a clenched fist. Additionally, grip strength decreases in the affected extremity, most notably in elbow extension, and is one of the few objective measures for diagnosing lateral epicondylitis. Dorf described testing grip strength while the elbow was flexed and extended. An 8% difference between flexion and extension grip strength was 83% accurate for diagnosing lateral epicondylitis.[11]

Pain with pressure directly over the radiocapitellar joint, especially when combined with active pronosupination of the forearm with axial loading in midrange elbow flexion (e.g., active radiocapitellar compression test), is indicative of radiocapitellar pathology.[12] In the middle-aged adult patient, point tenderness at the posterolateral elbow points more to radiocapitellar arthritis, although it may be more indicative of an OCD lesion in adolescent patients. These patients often have an effusion over the radiocapitellar joint.

Next, the "soft spot" over the anconeus can be palpated at the center of a triangle drawn between the olecranon, lateral epicondyle, and radial head. An effusion in that area can present with fullness over the soft spot and should raise suspicion for an underlying injury.

Local, vague tenderness in the region of the proximal lateral forearm, especially along the course of the PIN, is more consistent with the diagnosis of RTS versus PIN syndrome and is explained in more detail later in this chapter.

Medial Elbow

Palpation of the medial elbow begins with the medial epicondyle. Tenderness associated with medial epicondylitis is often located on the anteromedial facet of the medial epicondyle where the pronator teres and flexor carpi radialis originate. The involvement of the pronator origin can be isolated by palpating the proximal third of the medial epicondyle while resisting forearm pronation. The flexor carpi ulnaris origin is isolated by palpating the distal third of the medial epicondyle with resisted flexion and ulnar deviation.

The medial (ulnar) collateral ligament is best palpated when the elbow is flexed to 50 to 70 degrees, which anteriorly translates the medial muscles. Tenderness associated with medial collateral ligament tear will be along the course of the ligament either at the origin on the medial epicondyle or at the insertion on the sublime tubercle of the proximal ulna. The ulnar nerve can be palpated directly beneath the medial epicondyle (as described later in this chapter).

Anterior Elbow

Palpation of the anterior elbow begins in the antecubital fossa, where multiple soft tissue structures can be palpated: the distal biceps tendon, brachioradialis, lacertus fibrosus, brachialis, and pronator teres. Distal biceps tendon pathology is the most common source of anterior elbow pain in the athlete, and the tendon is readily palpable in the antecubital fossa with the elbow supinated. The hook test or biceps squeeze test can be performed to assist in the diagnosis of distal biceps tendon rupture; these tests are detailed later in this chapter.[13,14] Tenderness along the distal extent of the tendon with resisted supination in the face of an intact tendon suggests pain from distal biceps tendinosis.

Median nerve compression neuropathies, specifically pronator syndrome, can cause anterior elbow and forearm pain with palpation. Deep palpation or the Tinel test of the median nerve in the antecubital fossa can elicit pain in the proximal forearm and elbow, but median nerve symptoms would also be expected distally in the hand (as discussed later in this chapter).

Posterior Elbow

Palpation of the posterior elbow involves palpating the triceps tendon and the olecranon process. When the elbow is flexed to approximately 30 degrees, the triceps is relaxed, allowing for easier access to the medial and lateral gutters of the posterior compartment. Fullness in these regions is consistent with an intra-articular inflammatory process. Tenderness on the medial or lateral aspect of the trochlea can be seen in association with spurs on the posterior aspect of the distal humerus. Pain with terminal extension can be seen with spurs on the tip of the olecranon process. Boxers can fracture the tip of the olecranon process, particularly in their nondominant elbow, during a jab. Tenderness from this type of fracture may be elicited at the apex of the olecranon process with the elbow slightly flexed.

Pain elicited along the posteromedial aspect of the olecranon in overhead-throwing athletes should immediately warrant a focused exam for posteromedial impingement syndrome. It is also important to carefully evaluate the integrity of the medial collateral ligament because there is an association between medial collateral ligament injury and posteromedial impingement syndrome. The extension impingement and arm bar test are sensitive physical exam maneuvers that can be used (as discussed later in this chapter).

A palpable defect in the triceps just proximal to the tip of the olecranon is indicative of triceps tendon rupture. Complete ruptures typically involve the posterior superficial portion of the tendon, which is made up of the long and lateral heads of the triceps. This leaves the deep anterior portion of the tendon, which is composed of the medial head of the triceps, intact.

Motion

The normal arc of flexion-extension ranges from 0 to 140 degrees, ±10 degrees.[15] Loss of terminal extension is common in athletes with elbow pathology. Greater variation is found in the pronation-supination arc than in the flexion-extension arc; however, approximately 75 degrees of pronation and 85 degrees of supination are considered to be within normal ranges.[15] The arc of motion required for daily activities ranges from 30 to 130 degrees of flexion-extension and 50 degrees of pronosupination in each direction.[16] Depending on the sport, loss of terminal extension may create a significant impairment. It is important to note at which point in the arc of motion that pain or mechanical symptoms occur because this information may provide clues to the diagnosis.

Neurologic Assessment

The neurologic examination includes an assessment of skin appearance and texture, muscle bulk, strength, sensation, reflexes, and provocative maneuvers for nerve compression. Loss of skin texture and abnormal sweat patterns on the finger pads are signs of significant sensory denervation. Similarly, loss of muscle bulk follows motor denervation or limb disuse.

Elbow flexion strength is best tested with the elbow flexed 90 degrees and the forearm in neutral rotation. The contribution of flexion strength from the brachialis is disproportionate compared with the biceps. The brachialis inserts onto the anterior surface of the coronoid, which causes its line of pull to remain constant regardless of the position of the hand or wrist. This is in contrast to the biceps, which attaches onto the radial tuberosity and then contributes to both flexion and supination. Additionally, the brachialis is positioned closer to the axis of the elbow joint increasing its mechanical advantage compared to the biceps.

Triceps strength is demonstrated by extension of the elbow with and without the aid of gravity and then against increasing resistance. To eliminate the aid of gravity, the patient should be asked to extend the elbow while in the supine position with the arm forward flexed to 90 degrees and the elbow in 90 degrees of flexion (Fig. 58.2). Triceps strength is assessed with resisted elbow extension starting with the elbow in 100 degrees of flexion, which will assess the lateral and middle heads of the triceps. To isolate the medial head of the triceps, the examiner should have the patient place his or her affected hand on the examiner's shoulder with the elbow extended. The examiner then places his or her hands over the patient's antecubital fossa and attempts to flex the patient's elbow while the patient resists the motion. This will isolate the medial head of the triceps.

Pronation and supination are best evaluated with the elbow at 90 degrees of flexion and the forearm starting in neutral

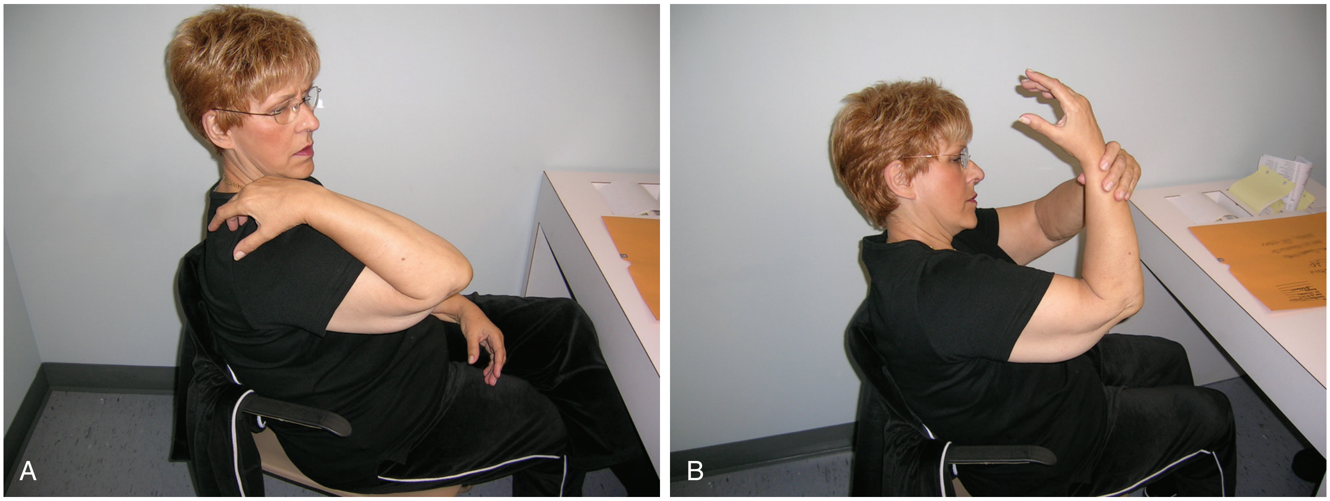

Fig. 58.2 Triceps insufficiency. Triceps function may be first demonstrated by having the patient attempt to extend the elbow against gravity. This maneuver is performed with the patient sitting and the humerus held perpendicular to the torso during elbow extension. The patient was unable to actively extend the elbow (A), although passive extension was possible (B), effectively indicating triceps insufficiency.

rotation. Although not an absolute rule, the dominant extremity has been shown to be approximately 10% stronger than the nondominant extremity.[17,18]

Sensation may be tested by lightly touching the patient in either the dermatomal or peripheral nerve distributions of the hand and forearm. Any decrease in light touch perception or difference from the contralateral side should raise suspicion for neurologic compromise.

COMPRESSION NEUROPATHIES AT THE ELBOW

Tests to detect potential compression neuropathy should also be performed when clinical suspicion is high based on the history and description of symptoms.

Ulnar Nerve Compression (Cubital Tunnel Syndrome)

Cubital tunnel syndrome is one of the most common compression neuropathies of the upper extremity, occurring in 1.8% of the population.[19] Various physical exam maneuvers can be performed to help with diagnostic decision-making. The Tinel test is performed by tapping a nerve (mixed peripheral or cutaneous) over a suspected area of compression or compromise (Fig. 58.3). The test is considered positive if paresthesias are produced in the distribution of the nerve in question.

The ulnar nerve is palpated posterior to the medial epicondyle as it travels through the cubital tunnel and distally into the flexor carpi ulnaris muscle mass. A painful snapping sensation can occur in two situations during elbow flexion: subluxation of the ulnar nerve and subluxation of the medial head of the triceps, with the latter typically occurring after ulnar nerve transposition. However, ulnar nerve hypermobility is present in up to one-third of the general population, with the vast majority of these individuals remaining asymptomatic from this phenomenon.[20]

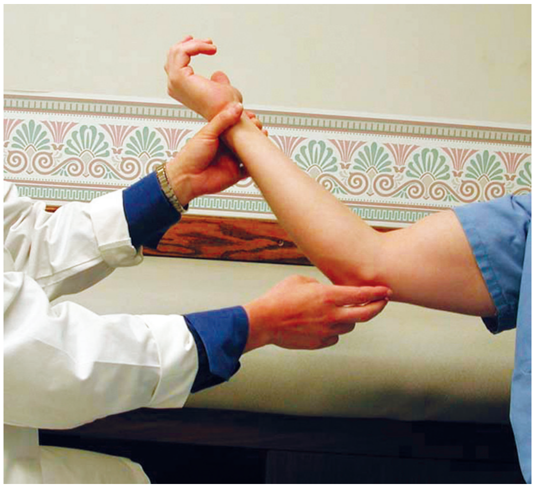

Fig. 58.3 The Tinel test is performed by tapping a nerve over the suspected area of compression or nerve irritation. The test is considered positive if it produces paresthesias in the areas of the nerve's typical distribution.

In addition to the Tinel test, ulnar nerve irritability may be elicited by prolonged maximal elbow flexion, direct pressure at the cubital tunnel, or a combination of the two (elbow flexion-pressure test). Multiple studies have evaluated the sensitivity, specificity, positive predictive value (PPV), and negative predictive value (NPV) for these maneuvers.[21-23] The sensitivity, specificity, PPV, and NPV for a Tinel test at the elbow ranges from 54% to 70%, 53% to 99%, 77% to 97%, and 30% to 98%, respectively.[21-23] The sensitivity, specificity, PPV and NPV for the elbow flexion-pressure maneuver ranges from 46% to 98%, 40% to 99%, 72% to 96%, and 29% to 99%, respectively.[21-23]

The scratch collapse test has also proved to be sensitive and accurate as a provocative maneuver for the diagnosis of cubital tunnel syndrome. In this test, the examiner scratches the patient's skin lightly over the area of nerve compression while the patient performs sustained resisted external rotation of the shoulder bilaterally.[23] Brief loss of muscle resistance will be elicited as a result of nerve-related allodynia. This test does not rely on the patient's report of symptoms but rather provides a more objective evaluation compared with most other examination maneuvers. The scratch collapse test has shown significantly higher sensitivity (69%) than the Tinel test (54%) and the elbow flexion-pressure test (46%) and has an overall accuracy approaching 90% for the diagnosis of cubital tunnel syndrome.[23]

Overall, provocative maneuvers are an essential clinical tool in assisting in the diagnosis of cubital tunnel syndrome. However, the examiner should still be aware that there can be up to a 34% false-positive rate, which was shown in a cohort of asymptomatic individuals with positive elbow flexion and the Tinel sign.[24]

Chronic ulnar nerve compression can result in the development of intrinsic muscle weakness and atrophy (commonly seen in the first dorsal web space). Many characteristic signs of chronic ulnar nerve compression can be found distally in the hand, including the "claw hand," and the Wartenberg, Froment, Masse, and Jeanne signs. The most notable of the chronic signs is clawing of the hand, which is synonymous with the terms *intrinsic minus position* and *the Duchenne sign* (Fig. 58.4). Nevertheless, all these descriptive eponyms describing paralysis of both the lumbrical and interosseous muscles point to an imbalance between the extensors and flexors of the digits. With loss of intrinsic muscle function to the digits, the extensor digitorum communis is unopposed at the metacarpalphalangeal joint, leading to extension at the joint. Additionally, the flexor digitorum superficialis and profundus are no longer opposed by the intrinsics, which causes proximal and distal interphalangeal joint flexion. Clawing is most commonly limited to the ring and small fingers and is associated with ulnar nerve lesions that are distal to the innervation of the flexor digitorum profundus to the fourth and fifth digits. Clawing in the index and long fingers is uncommon because of the median nerve's contribution to the lumbrical muscles of those digits. The Masse sign is a flattening of the palmar arch due to hypothenar muscle paralysis and results in a loss of tone across the transverse metacarpal arch. The Wartenberg sign presents as a fixed ulnar deviation and extension of the small finger at rest, with weakness of active adduction. This is secondary to unopposed extensor digiti quinti function, or a weak third palmar interosseous muscle. The Froment sign is a compensatory flexion by the flexor pollicis longus at the interphalangeal joint during key pinch secondary to a weak adductor pollicis brevis. If there is an associated hyperextension of the thumb metacarpophalangeal joint with weakness of thumb adduction, it is called the Jeanne sign.[25]

Fig. 58.4 Intrinsic minus or clawing of hand. This condition is caused by compression of the ulnar nerve leading to paralysis of the interossei muscles of the hand and the ulnar two lumbricals. The extensor digitorum comminus (EDC) is then unopposed at the metacarpophalangeal (MCP) joint causing MCP joint extension. The flexor digitorum superficialis (FDS) and flexor digitorum profundus (FDP) are also unopposed leading to flexion at the proximal interphalangeal (PIP) and distal interphalangeal (DIP) joints. Clawing caused by ulnar nerve compression is generally limited to the ring and small fingers as the first and second lumbricals are innervated by the median nerve.

Radial Nerve Compression Neuropathies

Radial nerve compression neuropathies that occur around the elbow include PIN syndrome and RTS.

Proximal compression of the superficial sensory branch of the radial nerve, also known as Wartenberg syndrome, occurs as the branch courses from a deep to superficial structure along the brachioradialis. Patients usually present with dorsoradial hand and forearm pain with sensory loss except in the elbow.

In assessing PIN compression syndromes, the physician must understand the clinical differences between PIN syndrome and RTS. RTS is a painful condition localized to the proximal lateral forearm, whereas PIN syndrome is a pure motor palsy. Although they are clinically different, both syndromes involve compression of the PIN and other common sites of compression.

Radial Tunnel Syndrome

Roles and Maudsley[26] described maneuvers that elicited pain in a cohort of 36 patients with RTS who were believed to have "resistant" tennis elbow. They most commonly provoked symptoms with resisted supination, wrist extension, and middle finger extension. Traction on the radial nerve can also be maximized by extending the elbow in pronation and flexing the wrist. Palpation of the radial tunnel, located 3 to 5 cm distal to the lateral epicondyle, can elicit pain. This can be performed by using the rule-of-nine test described by Loh,[27] which consists of drawing a square over the anterior portion of the proximal forearm (Fig. 58.5) measured as the width of the elbow crease with the elbow fully extended and fully supinated. This measured width will then serve as the length distally as well. The square is then further subdivided into nine squares, with three columns and three rows. The nine squares are then numbered one to three from proximal to distal in the lateral, middle, and medial columns. The authors found that the nerve was present in the proximal two boxes in the lateral column on all specimens studied. Therefore tenderness over the lateral boxes during palpation indicates irritation consistent with RTS. The other boxes serve as controls, and no tenderness should be noted over these areas.

To help distinguish RTS from lateral epicondylitis, a local anesthetic and/or corticosteroid may be injected along the course

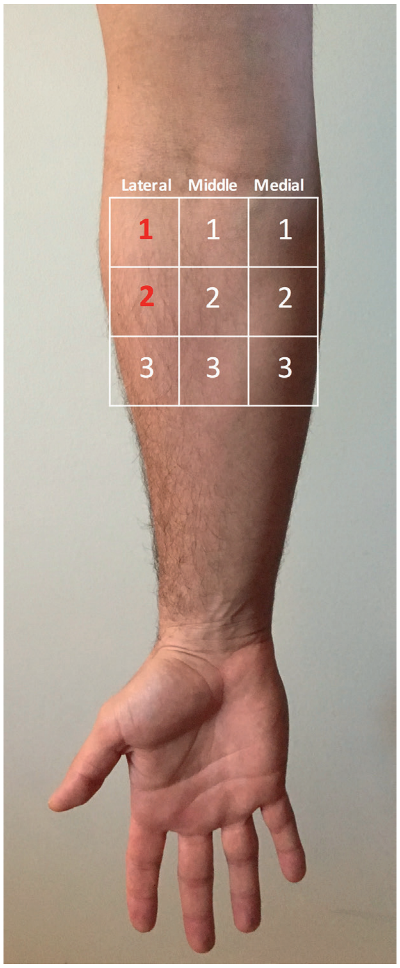

Fig. 58.5 The rule-of-nine test for radial tunnel syndrome. The anterior forearm is subdivided into nine squares. The lateral, middle, and medial columns are divided into three squares numbered proximally to distally. Patients with radial tunnel syndrome will note tenderness over squares one and two in the lateral column where the radial nerve resides in the proximal forearm. The other squares serve as controls.

of the PIN in the proximal forearm approximately 3 to 5 cm distal to the lateral epicondyle. This is more distal than the common source of pain associated with lateral epicondylitis. Despite the justified hesitation of most practitioners, it is recommended that enough lidocaine be administered to provide a temporary PIN motor block to confirm correct placement of the injection.

Sarhadi et al.[28] reported on 25 patients with suspected RTS, 16 of whom showed pain relief at 2 years with an injection of 1% lidocaine (which initially resolved symptoms in all patients) and 40 mg of triamcinolone for long-term pain control. Nonetheless, it is not our practice to inject an anesthetic or steroid in the vicinity of the posterior interosseous nerve for the purpose of diagnosis or treatment. For baseball pitchers, provocative maneuvers that can increase tension on the PIN include elbow extension, forearm pronation, and wrist flexion. This position occurs at the end of ball release; thus reproducing pain in the lateral proximal forearm in this position in the office setting may lend more weight toward a working diagnosis of RTS, especially if there is no motor loss.

Posterior Interosseous Nerve Syndrome

PIN syndrome is distinct from RTS in that patients with PIN syndrome have a significant motor deficit. This includes loss of finger and thumb extension and radial deviation of the wrist during extension because of the consistent PIN innervation of the extensor carpi ulnaris as opposed to the extensor carpi radialis longus (which is usually innervated by the radial nerve proper).[29] This condition may result from atypical (deep) lipomas, inflammatory synovitis, or a local hematoma from trauma. Temporary and even permanent dysfunction of the PIN can follow neurolysis of the PIN during exposure for fracture fixation and surgical decompression for RTS. PIN palsy is also a recognized complication of surgery for a distal biceps tendon rupture, occurring in 2.7% of one-incision approaches and in 0.2% of two-incision approaches.[30]

It is also possible to experience partial or complete PIN dysfunction in association with brachial neuritis, or Parsonage-Turner syndrome. Partial lesions may only involve the medial branch of the PIN, causing weakness of the extensor carpi ulnaris, extensor digiti minimi, and extensor digitorum communis, whereas involvement of the lateral branch can cause weakness of the abductor pollicis longus, extensor pollicis brevis, extensor pollicis longus, and extensor indicis proprius.[31] A thorough and accurate exam is therefore essential to localize lesions that may present with a variable clinical picture.

Median Nerve Compression at the Elbow

Pronator Syndrome

Pronator syndrome is a rare clinical entity that is thought to result from compression of the median nerve as it passes through the two heads of the pronator teres or the proximal aspect of the flexor digitorum superficialis in the proximal forearm. Patients present with volar proximal forearm pain and paresthesias of the first three digits and radial half of the index finger. These symptoms can be exacerbated by repetitive physical activities. Discriminating between pronator syndrome and carpal tunnel syndrome is important and based on the anatomic relationship of the palmar cutaneous branch, which arises 4 to 5 cm proximal to the transverse carpal ligament.[32] A patient with a more proximal compression lesion of the median nerve will have thenar eminence numbness versus a true carpal tunnel compression, which presents with symptoms in the first three digits and radial half of the ring finger and normal sensation of the thenar region. Anterior interosseous nerve syndrome can be differentiated from both carpal tunnel and pronator syndrome because it presents with only motor symptoms, primarily of the flexor pollicis longus and flexor digitorum profundus of the index finger and no loss of sensation.

The provocative maneuvers for pronator syndrome vary. Irritation of the nerve at the level of the pronator teres can be elicited by resisted pronation of the forearm while moving the elbow from flexion to extension.[33,34] Other maneuvers include resisted flexion of the flexor digitorum superficialis to the middle finger and resistance of the forearm in almost full flexion and full supination, which are thought to result from compression at the level of the fibrous arch between the heads of the flexor digitorum superficialis and lacertus fibrosus, respectively.[33-35] Additionally, a positive scratch collapse test and Tinel test may be observed in patients with proximal median neuropathy. The diagnosis of pronator syndrome is sufficiently rare that the accuracy of these provocative maneuvers is based more on expert opinion than on evidence.

Anterior Interosseous Nerve Syndrome

Compression of the AIN is a motor-only palsy associated with vague proximal forearm pain and significant motor deficits of the flexor pollicis longus, flexor digitorum profundus to the index and middle fingers, and pronator quadratus. Having the patient make an "OK" sign while paying specific attention to whether he or she is flexing the thumb interphalangeal joint can quickly assess AIN function. Isolating pronator quadratus function is much more difficult because the pronator teres function overpowers its isolated function. Flexing the elbow maximally can keep the humeral head of the pronator teres from contributing to forearm pronation; however, this still completely isolates pronator quadratus function.

JOINT STABILITY TESTING

Lateral Collateral Ligament

Most patients with recurrent elbow instability secondary to a history of previous trauma (e.g., dislocations or fracture-dislocations of the elbow) or surgical procedures such as tennis elbow surgery can have posterolateral rotatory instability resulting from a deficient lateral collateral ligament complex. Patients will report chronic instability with common daily maneuvers and possibly frank dislocation.

A varus stress test can be used to assess direct stability of the lateral side with the elbow at 30 degrees of flexion. However, the most commonly used test to assess for posterolateral rotary instability is the lateral pivot-shift test. This test was first described by O'Driscoll[36] and is performed with the patient relaxed in the supine position. With the examiner at the patient's head, the forearm is supinated while a valgus stress with an axial load is applied to the elbow. The elbow is then slowly extended from full flexion. The test is considered positive when posterolateral subluxation of the radial head occurs at approximately 40 degrees of extension. Flexion from that point reduces the subluxed radial head. This maneuver produces subluxation 38% of time only in a patient who is awake. Therefore an examination with use of general anesthesia is warranted when the history suggests recurrent posterolateral instability with a positive maneuver elicited in 87.5% to 100% of patients.[37,38]

The combination of elbow extension and forearm supination may elicit a sense of apprehension in an awake patient. In a study of eight patients with posterolateral rotatory instability, the use of a ground or chair push-up test was compared with the lateral pivot-shift test.[37] The push-up test is a maneuver that can be done on the ground or while sitting and involves supination with the elbows flexed to 90 degrees and arms greater than shoulder width apart. A positive test was determined as either guarding or dislocation as the elbows are extended during each maneuver. Although these tests were more sensitive than the lateral pivot-shift test in awake patients, there was no comparison to a negative control group. Performing the push-up test may therefore confound the physician for other potential causes of elbow pain and instability if pain is the only complaint.

Medial (Ulnar) Collateral Ligament

Medial (ulnar) collateral ligament instability is evaluated by numerous tests, including the valgus stress test, the "milking maneuver," and the moving valgus stress test.

The valgus stress test is performed with the shoulder held in abduction and external rotation. The examiner stabilizes the arm and flexes the elbow to 30 degrees to unlock the ulnohumeral joint. A valgus stress is then placed on the elbow. The test is positive if the patient has pain, apprehension, or gross laxity of the medial side. Sensitivity and specificity vary depending on whether the patient has pain with the maneuver versus laxity. Pain elicited by the exam has higher sensitivity (65%) versus laxity (19%) but has lower specificity (50%) versus laxity on exam (100%).[39]

The milking maneuver is performed by extending and externally rotating the humerus to restrict glenohumeral motion. The forearm is then supinated and a valgus stress is applied by pulling on the extended thumb with the elbow flexed at 90 degrees. This can be done by the patient or the examiner. The ligament is palpated during the stress maneuver, and the test is positive if there is pain, apprehension, or gross medial laxity (Fig. 58.6). Attention should be paid to baseball pitchers because they can have increased laxity of the medial collateral ligament of the dominant arm.[40] What is perceived as laxity of the medial collateral ligament can therefore actually be a false-positive result in what is otherwise a normal variant for these athletes.

The moving valgus stress test, developed by O'Driscoll, is a dynamic milking maneuver that addresses concerns regarding the position of the elbow and the sensitivity of a static valgus stress test, arthroscopic valgus stress test, and MRI in the diagnosis of

Fig. 58.6 The "milking maneuver" test. This is performed by extending and externally rotating the humerus to restrict glenohumeral motion. The forearm is then supinated and a valgus stress is applied by pulling on the extended thumb with the elbow flexed at 90 degrees. The test is considered positive if it elicits pain.

Fig. 58.7 The moving valgus stress test is performed with the shoulder held in abduction and external rotation. The examiner stabilizes the arm and flexes and extends the elbow through an arc of motion. The test is considered positive if the medial elbow pain is reproduced at the ulnar collateral ligament.

medial collateral ligament tears. The patient's arm is placed in the same position as in the milking maneuver but a constant valgus stress is placed at the elbow as it is moved through an arc of flexion and extension (Fig. 58.7). Based on analysis, the moving valgus stress test has a high sensitivity (100%) and specificity (75%) for ulnar collateral ligament injury.[39] The test is considered positive if it reproduces the patient's symptoms between a range of 70 to 120 degrees. The mean angle to reproduce the most pain in patients was 90 degrees.[39]

OTHER PROVOCATIVE MANEUVERS

Distal Biceps Tendon Rupture

For distal biceps tendon ruptures, specific physical exam maneuvers can help to assess the integrity of the tendon at its insertion site. O'Driscoll et al.[13] have described the hook test and Ruland et al.[14] reported on the use of the biceps squeeze test.

The hook test involves the examiner hooking his or her index finger under the intact biceps tendon from the lateral side while the patient fully supinates the forearm and flexes the elbow to 90 degrees. When performing the test, it is beneficial to have the patient make the "thumbs up" sign during supination and hold the arm in almost 90 degrees of abduction. The examiner can also approach the tendon from the medial side underneath the lacertus fibrosus. This is more difficult, however, because the finger cannot be hooked underneath as far. If the examiner can hook his or her finger under the tendon, then it is intact. An abnormal test corresponding to an avulsed tendon would indicate an inability to hook the finger underneath the tendon. In the study by O'Driscoll et al.,[13] the hook test was compared with MRI and intraoperative findings of patients with a distal biceps tendon rupture. The sensitivity and specificity were both 100% with the hook test compared with 92% and 85% for MRI alone.[13]

The biceps squeeze test is similar in concept to the Thompson test for Achilles tendon rupture. The biceps muscle belly is squeezed with one hand while the examiner's other hand squeezes the musculotendinous junction. The patient's arm should be resting at approximately 60 to 80 degrees of flexion. A normal exam would demonstrate forearm supination when squeezing the muscle belly and tendinous junction. A positive test would not demonstrate supination of the forearm. Based on one patient population, this test had a reported sensitivity of 95%, and all patients who had surgery had return of forearm supination when assessed with the biceps squeeze test.[14]

Posteromedial Impingement Syndrome

The extension impingement and arm bar tests are useful for evaluating posterior impingement in athletes secondary to posteromedial impingement syndrome. The extension impingement test is performed by palpating the posteromedial aspect of the olecranon with the patient's arm near full extension at the elbow (20 to 30 degrees) and then quickly extending the elbow fully to terminal extension with a continuous valgus stress. The purpose of this maneuver is to recreate the terminal phase of a valgus extension overload typically seen in overhead-throwing athletes. The valgus stress is necessary to identify whether the pain is primarily from posteromedial impingement and not from instability or other sources of pain. The test can also be performed without valgus or in varus and appropriate comparisons can be made.[41]

The arm bar test is performed with the arm in full internal rotation at 90 degrees of forward flexion with the hand on the examiner's shoulder. The patient's olecranon/distal humerus is pulled down to recreate full extension. A positive test will elicit pain in the patient's posteromedial elbow. This is considered a more sensitive test than the extension impingement test. It therefore may not be unusual to examine a pitcher in the recovery phase who still has a positive arm bar test but a negative extension impingement exam.[42]

CONCLUSION

The elbow is a complicated joint; it can be affected by many associated disorders in the athlete. Therefore a focused history and physical examination of the lateral, medial, anterior, and posterior aspects of the elbow can help the physician to develop a narrow differential diagnosis and guide treatment decisions that will allow the patient's return to play in a timely and safe manner.

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For a complete list of references, go to ExpertConsult.com.

SELECTED READINGS

Citation:

Cheng C, Mackinnon-Patterson B, Beck JL, et al. Scratch collapse test for evaluation of carpal and cubital tunnel syndrome. *J Hand Surg Am*. 2008;33A:1518–1524.

Level of Evidence:
II

Summary:
This study evaluates the usefulness of a new physical examination test for carpal tunnel syndrome and cubital tunnel syndrome. This test had a higher sensitivity (62%) compared with the Tinel test (32%) and the wrist flexion/compression test (44%) in diagnosing carpal tunnel syndrome. This test also had a higher sensitivity (69%) than the Tinel test (54%) and the elbow flexion/compression test (46%) in diagnosing cubital tunnel syndrome. The overall conclusion is that this physical examination maneuver is highly sensitive and useful in the diagnosis of carpal and cubital tunnel syndrome.

Citation:
Rosenthal EA. Examination of hand and forearm: claw hand deformity & thumb deformity with ulnar palsy. In: Peimer CA, ed. *Surgery of the Hand and Upper Extremity*. New York: McGraw-Hill; 1996:83–87.

Level of Evidence:
V

Summary:
This book chapter details the features of various nerve palsies and how they may manifest in the hand.

Citation:
O'Driscoll SW, Lawton RL, Smith AM. The "moving valgus stress test" for medial collateral ligament tears of the elbow. *Am J Sports Med*. 2005;33:231–239.

Level of Evidence:
II

Summary:
This study evaluates a physical examination maneuver for medial collateral ligament (MCL) injuries of the elbow. The moving valgus stress test was 100% sensitive and 75% specific in assessment of medial elbow pain from MCL injuries compared with other standards of diagnosis. The moving valgus stress test is another highly sensitive physical examination maneuver that can aid clinicians in narrowing the differential diagnosis of medial elbow pain with specific consideration of an MCL injury.

Citation:
O'Driscoll SW, Goncalves LB, Dietz P. The hook test for distal biceps tendon avulsion. *Am J Sports Med*. 2007;35(11):1865–1869.

Level of Evidence:
II

Summary:
This study evaluates the usefulness of the hook test when assessing a potential distal biceps tendon rupture. The hook test was found to have a sensitivity and specificity of 100% for a distal biceps tendon rupture, which is higher than magnetic resonance imaging (92% and 85%, respectively).

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